

KARAN KUMAR

Dhanbad, Jharkhand, India • +91 9608439061 • Karankumar.g870@gmail.com • [linkedin.com/in/karan-kumar](https://www.linkedin.com/in/karan-kumar)

CAREER OBJECTIVE

Pre-final year Computer Science undergraduate with strong fundamentals in object-oriented programming, database management, and full-stack development. Proficient in building scalable, data-driven enterprise applications using OOP principles, SQL databases, and modern web technologies (JavaScript, HTML5, CSS3). Seeking a Software Engineer role where I can leverage my technical foundation, fast-learning ability, and product mindset to contribute to cutting-edge enterprise software solutions in a high-velocity development environment.

TECHNICAL SKILLS

Programming Languages & OOP: C++ (OOP), Java (OOP), Python, JavaScript, SQL (MySQL, MS SQL familiar)

Web Technologies: HTML5, CSS3, JavaScript, ReactJS, REST APIs

Backend & Database: MySQL (proficient), MongoDB (familiar), SQL database design and optimization

Core CS & Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, System Design (basic)

PROJECTS

Personal Finance & Transaction Tracker — Python & MySQL

Dec 2025 – Feb 2026

- Designed and developed an enterprise-grade financial transaction system with multi-account management, categorized spend analytics, and rule-based fraud detection using OOP principles and MySQL database
- Built scalable database schema with normalized tables, implemented data validation logic, and created comprehensive unit tests — demonstrating full-stack ownership and enterprise application development skills

Inventory Management System — Python & MySQL

March 2025 – May 2025

- Engineered a production-ready inventory management application with complete CRUD operations, real-time reporting, and data audit trails using object-oriented design patterns
- Implemented RESTful data flow architecture, MySQL database with triggers and stored procedures, and comprehensive error handling — ensuring system reliability and data integrity

Placement Management System — C++ (OOP)

Sept 2024 – Dec 2024

- Architected a scalable placement management system using advanced C++ OOP concepts (inheritance, polymorphism, encapsulation) and core data structures
- Delivered complete technical documentation, design specifications, and peer-reviewed code following software engineering best practices — prepared for enterprise development workflows

AI-Driven Crop Production Analysis — Python

Oct 2024 – Jan 2025

- Developed a data processing pipeline on real-world datasets using Python and AI models — delivered production-quality code with comprehensive documentation in an Agile team environment, demonstrating cross-functional collaboration

EDUCATION

Chandigarh University — Bachelor of Engineering in Computer Science & Engineering

Expected Graduation: 2027 | CGPA: 7.29/10 | Location: Mohali, Punjab

Pre-University: Class XII: 86% | Class X: 89% | Rajkamal Saraswati Vidya Mandir School, Dhanbad, Jharkhand

ACHIEVEMENTS & PROFESSIONAL DEVELOPMENT

- Tekathon 2025 Participant — delivered a functional software product under competitive pressure; demonstrated product thinking, rapid prototyping, and team collaboration in a hackathon environment
- Active Competitive Programmer — working on consistent solving algorithmic problems on LeetCode and Codeforces; strong analytical and problem-solving skills with proven debugging capabilities
- Certified in Web Technologies — completed Coursera certifications in Python, JavaScript, HTML5, and CSS3 (University of Michigan); self-driven continuous learner committed to technical excellence
- Research Contribution — co-authoring an AI research paper; demonstrates technical depth, academic rigor, and research-driven engineering mindset
- Multilingual Professional — fluent in Hindi, Bangla, and English; effective communicator across diverse, cross-functional teams
- Strong OOP fundamentals with C++, Java, and Python and proven track record of building database-driven applications with MySQL